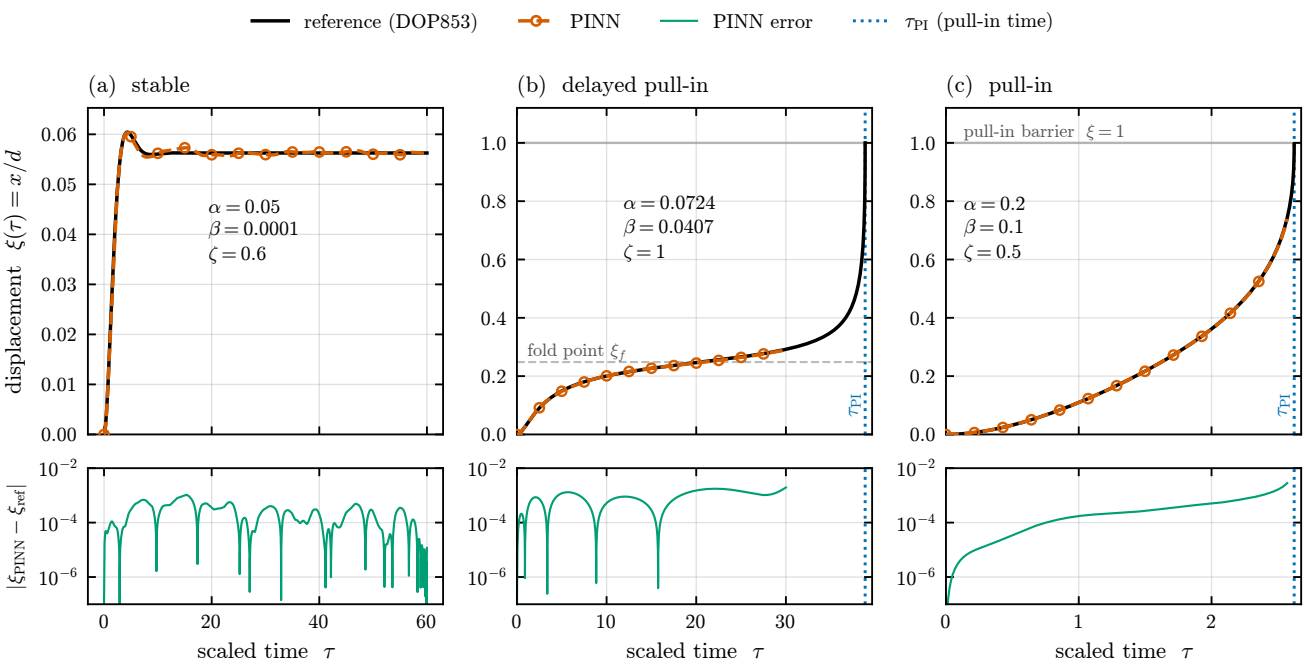
reference (DOP853)
PINN
PINN error
τ_{PI} (pull-in time)
(a) stable
(b) delayed pull-in
(c) pull-in
displacement $\xi(\tau) = x/d$
$|\xi_{\mathrm{PINN}} - \xi_{\mathrm{ref}}|$
$\alpha = 0.05$
$\beta = 0.0001$
$\zeta = 0.6$
$\alpha = 0.0724$
$\beta = 0.0407$
$\zeta = 1$
fold point ξ_f
pull-in barrier $\xi = 1$
$\alpha = 0.2$
$\beta = 0.1$
$\zeta = 0.5$
τ_{PI}
scaled time τ
scaled time τ
scaled time τ